

Population genomic insights of the Indo-Burmese pangolin (*Manis indoburmanica*)

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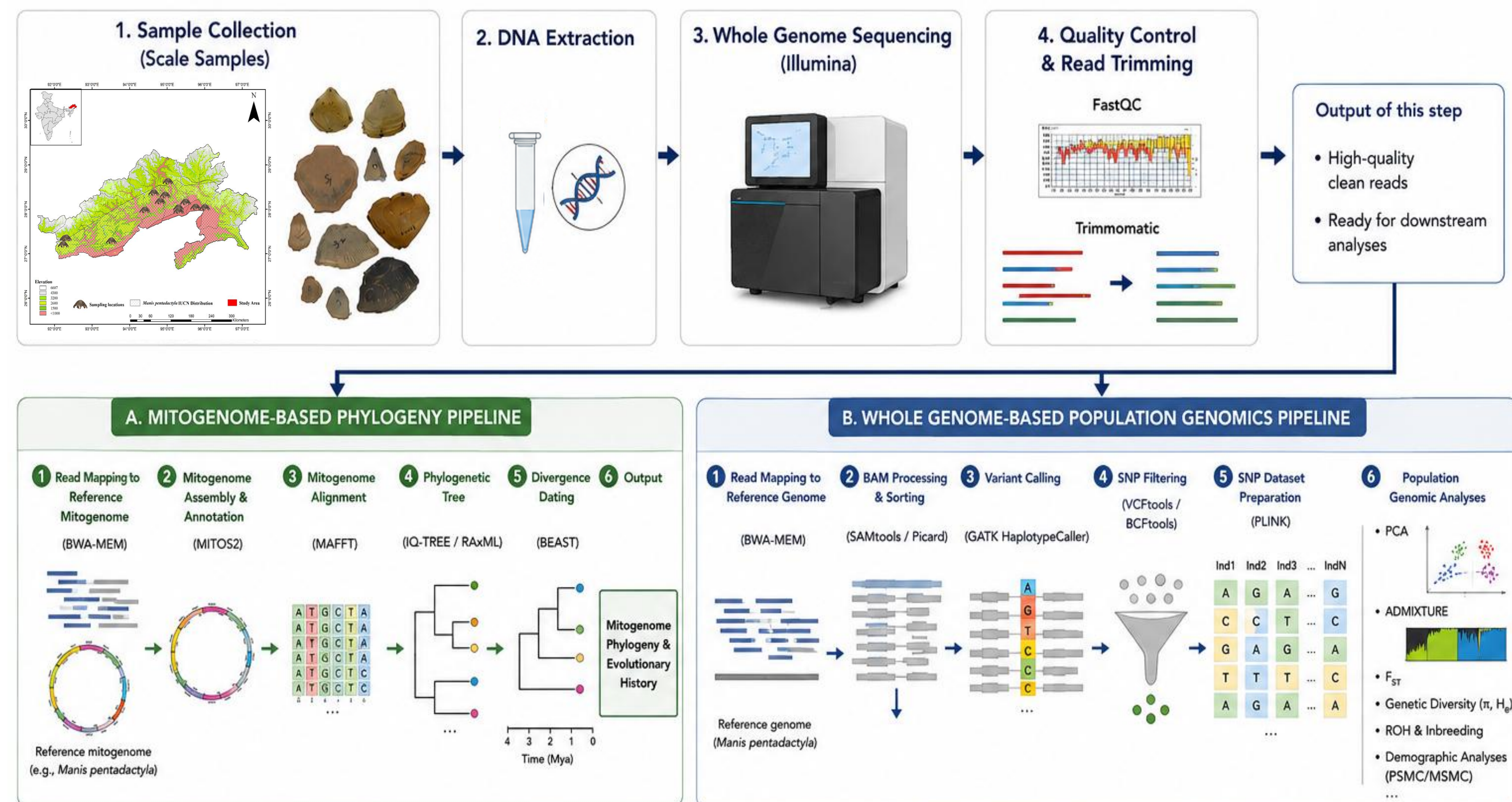
BACKGROUND AND INTRODUCTION

- Pangolins are the most trafficked mammals on earth.
- Asian pangolins (*Manis* spp.) exhibit significant cryptic diversity, with recent discoveries revealing evolutionary lineages.
- Quaternary and Pleistocene climatic shifts shaped pangolin diversification, distribution, and population history across Asia.
- Recent phylogenetic analyses identified a distinct lineage from Northeast India, described as the Indo-Burmese pangolin (*Manis indoburmanica*) diverged from *Manis pentadactyla* approximately 3.4 million years ago.

OBJECTIVES

- Investigate the evolutionary history of the Indo-Burmese pangolin (*Manis indoburmanica*) using genomic data.
- Assess the influence of geological and climatic changes on its divergence and distribution.

METHODOLOGY



RESULTS

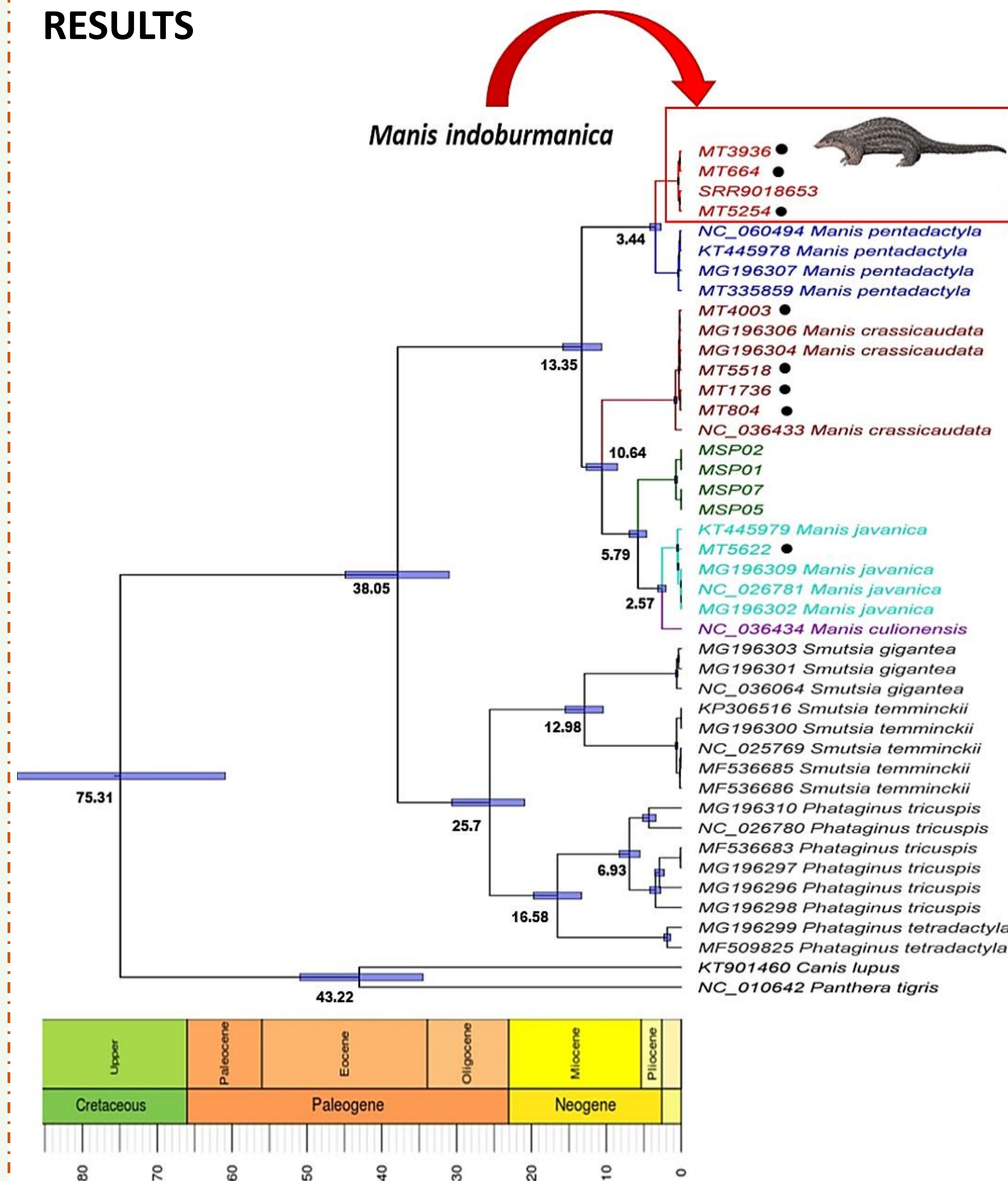


Fig. a) Time-calibrated Bayesian tree reconstructed from mitochondrial genomes, revealing the distinct Indo-Burmese pangolin (*Manis indoburmanica*) lineage and its divergence from the Chinese pangolin (*Manis pentadactyla*) approximately 3.4 mya.

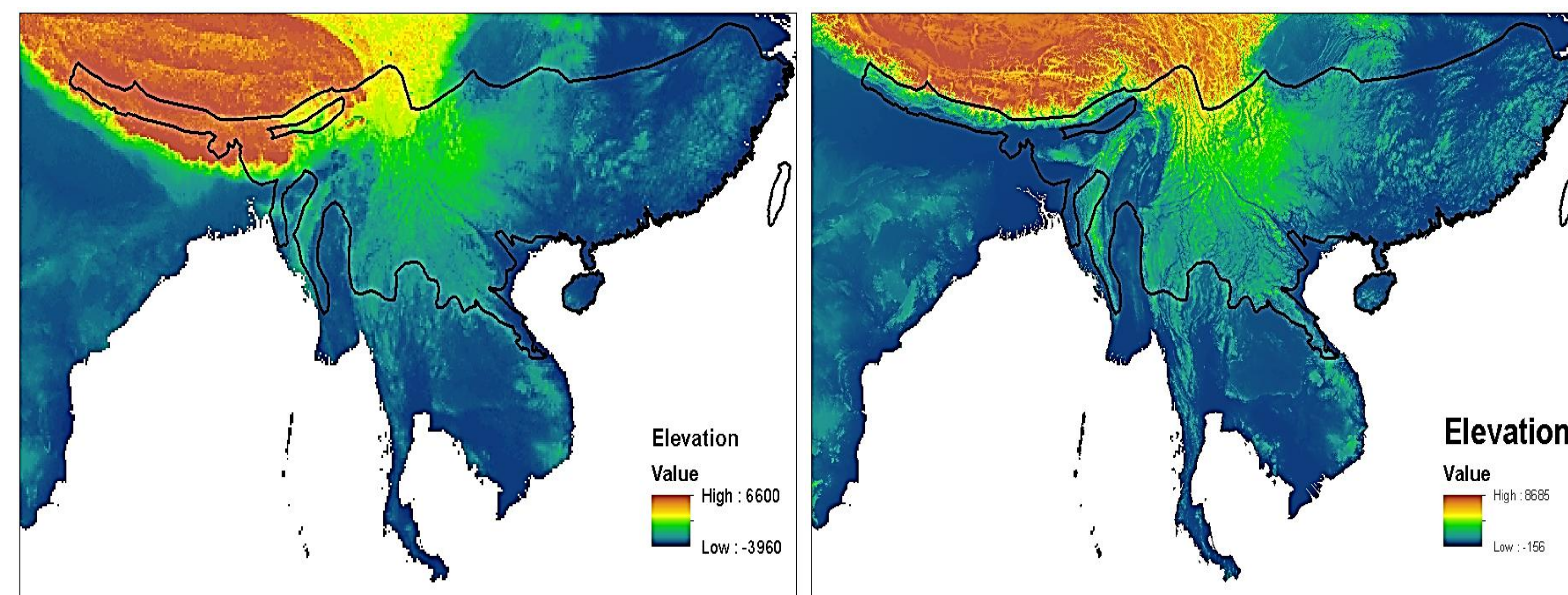


Fig. b) PaleoDEM model of the Eastern Himalaya showing topographic changes from ~5 mya to the present that may have contributed to the divergence of *Manis indoburmanica*.

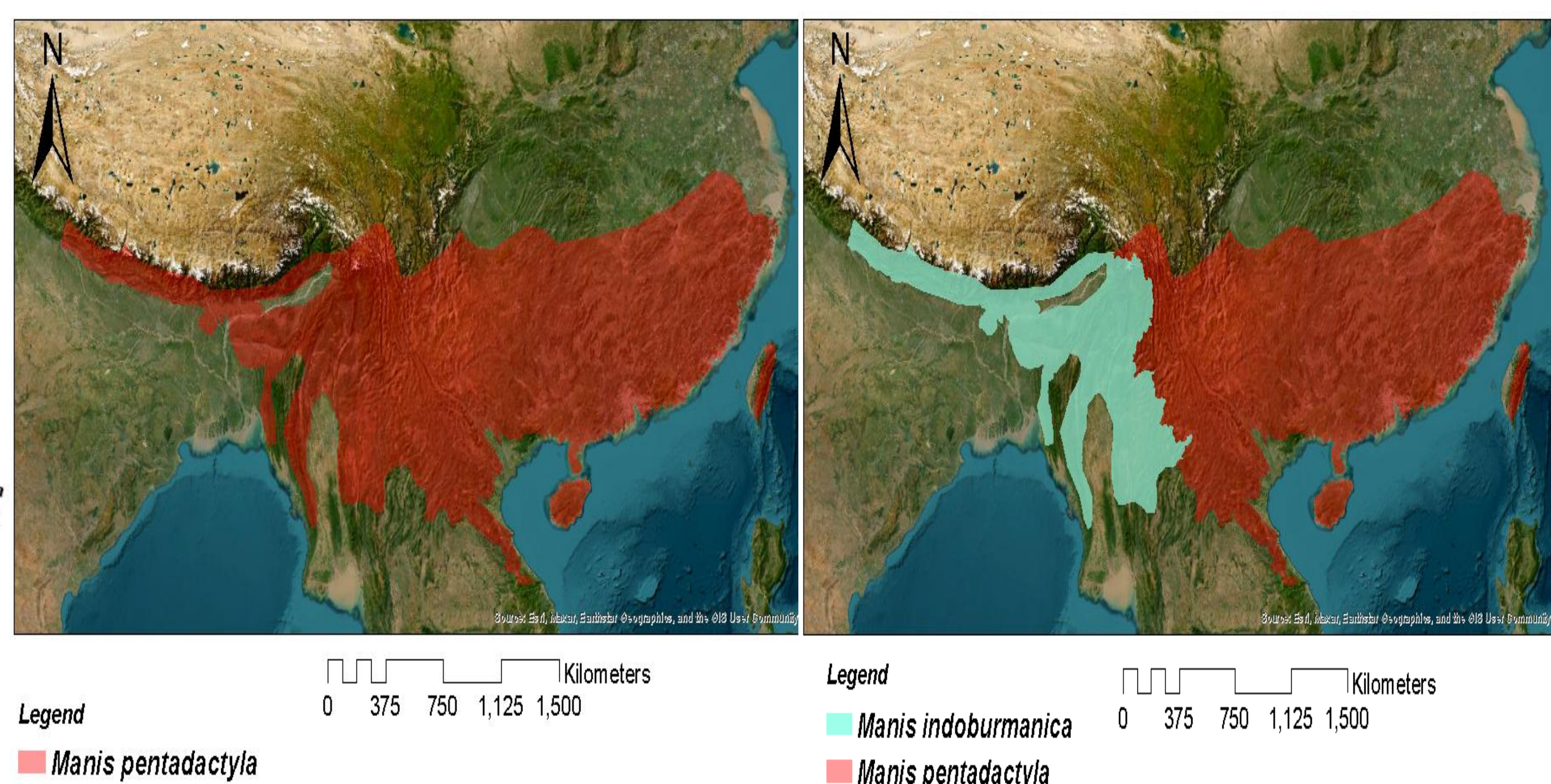


Fig. c) IUCN distribution of the Chinese pangolin (*Manis pentadactyla*) (left) and the putative distribution of the Indo-Burmese pangolin (*Manis indoburmanica*) (right), illustrating the proposed geographic range of the newly recognized lineage.

KEY FINDINGS

- Identified a distinct lineage of Asian pangolin, Indo-Burmese pangolin (*Manis indoburmanica*) diverged from Chinese pangolin (*Manis pentadactyla*) approximately 3.4 mya.
- Geological uplift and climatic fluctuation may have contributed to evolutionary divergence.
- A putative geographic range was delineated for *Manis indoburmanica*.

FUTURE DIRECTIONS

- Assess population structure and genetic diversity using genome-wide SNPs.
- Investigate gene flow, inbreeding and demographic history across the population of *Manis indoburmanica*.
- Identify conservation units to support species specific management and conservation planning.

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CONNECT WITH ME



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